ON THE SET FUNCTION IN $\mathbb{R}^n$
MEASURE AND INTEGRAL

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Let u be a nonnegative set function on a σ-algebra Σ (or algebra, or just any family of sets with some simple structure) in $\mathbb{R}^n$, i.e. $u : \Sigma \to [0, +\infty]$. The above mentioned Σ will most likely contain all the close and open balls/cubes.

Examples (i) u is an outer measure and Σ is just all subsets of $\mathbb{R}^n$.

(ii) $u[a, b] = f(b) - f(a)$ where f is nondecreasing on an interval I in $\mathbb{R}$. Note that this induces an outer measure on Σ if it is the collection of all subset of I. Note that it is additive on disjoint union of closed intervals.

(iii) u is a measure.

We will construct the smallest set of assumptions on u such that the following limit converges for almost every x in some set $\Omega \subset \mathbb{R}^n$:

$$Du(x) = \lim_{B \searrow x} \frac{u(B)}{|B|}$$

where $B \searrow x$ denotes the contraction of B into a singleton x, and $|B|$ denotes the Lebesgue measure of B.

There are two typical cases of contractions that we will be using.

(i) closed cubes/balls with center x with radius/edgelength tends to 0.

(ii) close cubes/balls that contain x with radius/edgelength tends to 0.

For simplicity, we will consider only case (i) where B is a closed ball of radius $r > 0$ centered at x, i.e. $B = \overline{B}_r(x)$. Define

- $\overline{D}u(x) = \limsup_{B \searrow x} \frac{u(B)}{|B|} = \lim_{r \to 0^+} \left(\sup_{t \in (0,r)} \frac{u(B_t(x))}{|B_t(x)|} \right)$
- $\underline{D}u(x) = \liminf_{B \searrow x} \frac{u(B)}{|B|} = \lim_{r \to 0^+} \left(\inf_{t \in (0,r)} \frac{u(B_t(x))}{|B_t(x)|} \right)$

The necessary and sufficient condition for the existence of $Du(x)$ is $\overline{D}u(x) = \underline{D}u(x)$. If $\overline{D}u(x) > \underline{D}u(x)$, let $\alpha, \beta \in \mathbb{Q}$ such that $\overline{D}u(x) > \alpha > \beta > \underline{D}u(x) \geq 0$. Thus $Du(x)$ exists for almost every $x \in \Omega$ (but it could be infinite) if

$$A_{\alpha,\beta} = \{x \in \Omega : \overline{D}u(x) > \alpha > \beta > \underline{D}u(x)\}$$

has measure zero for all α, β in $\mathbb{Q}$ (however, we will just show that $|A_{\alpha,\beta}| = 0$ for all $\alpha > \beta$).

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Observe that if $x \in A_{\alpha\beta}$, then given any $\delta > 0$, there exist balls (or cubes depending on our choice) $B^{(\alpha)}, B^{(\beta)}$ centered at x (or containing x for case (ii)) with radius less than δ such that $\frac{u(B^{(\alpha)})}{|B^{(\alpha)}|} > \alpha > \beta > \frac{u(B^{(\beta)})}{|B^{(\beta)}|}$. That is, the collections $\{B^{(\alpha)}\}$ and $\{B^{(\beta)}\}$ each cover $A_{\alpha,\beta}$ finely.

Lemma 0.1 (Vitali Covering Theorem for Lebesgue measure)**.** *A collection $\mathcal{V}$ of closed balls is said to cover a set E finely if for every $x \in E$ and $\eta > 0$, there is a ball/cube B containing x whose radius/edgelength is less than η. Suppose E is covered finely by a collection $\mathcal{V}$ of balls/cubes such that $0 < |E|_e < +\infty$ [1]. Then, given any $\epsilon > 0$, there is a countable collection M of disjoint balls in $\mathcal{V}$ such that*

$$|E \setminus \bigcup_{B_k \in M} B_k|_e = 0 \text{ and } \sum_{B_k \in M} |B_k| < |E|_e + \epsilon.$$

Consequently, there exists a finite sequence of pairwise disjoint balls $\{B_k\}_{k=1}^N$ in $\mathcal{V}$ such that

$$|E \setminus \bigcup_{k=1}^N B_k|_e < \epsilon, \ \sum_{k=1}^N |B_k| < |E|_e + \epsilon \ \text{ and } |E \cap (\bigcup_{k=1}^N B_k)|_e \geq |E|_e - \epsilon.$$

Let us make the following two assumptions

Assumption 0.2 (Additivity for disjoint closed balls/cubes)**.**

$$A, B \in \Sigma \text{ and } A \cap B = \emptyset \implies A \cup B \in \Sigma \text{ and } u(A \cup B) = u(A) + u(B)$$

Assumption 0.3 (monotoncity)**.**

$$A, B \in \Sigma \text{ and } A \subseteq B \implies u(A) \leq u(B).$$

The above two assumptions holds for Borel measures and outer measures.

Before we begin, first note that as derivatives are local properties, it is enough to consider just bounded set Ω.

Claim 0.4. *For any open set U that contains a bounded subset $E \subseteq A_{\alpha,\beta}$, we have $u(U) \geq \alpha|E|_e$.*

Since U is open, for each $x \in E$, we can find a subcollection of balls $\{B_U^{(\alpha)}\} \subseteq \{B^{(\alpha)}\}$ that finely covers E and is contained in U. By Vitali Covering Theorem for Lebesgue measure on finite outer measure E with $\{B_U^{(\alpha)}\}$, there exists pairwise disjoint balls $\{B_i\}$ contained in U such that $|E \setminus \bigcup B_i|_e = 0$ with $u(B_i) > \alpha|B_i|$ for all i.

[1] if $|E|_e = 0$, the result is immediate

Then, for all $N \in \mathbb{N}$,

$$\begin{aligned} u(U) &\geq u\left(\bigcup_{i=1}^N B_i\right) && \text{(monotoncity)} \\ &= \sum_{i=1}^N u(B_i) && \text{(additivity)} \\ &> \alpha \sum_{i=1}^N |B_i| && . \end{aligned}$$

As the inequality is true for all N, $u(U) \geq \alpha \sum_{i=1}^\infty |B_i| \geq \alpha |E|_e$.

Remark Indeed, we could also show that $\overline{D}$ is finite a.e. if u is locally finite. This is because the set

$$A_\infty \{x \in A : \overline{D}u(x) = \infty\} \text{ is a subset of } A_\alpha = \{x \in A : \overline{D}u(x) > \alpha\}$$

for all $\alpha > 0$. Taking U as any open set containing A_∞ such that $u(U) < \infty$, we know that $u(U) > \alpha |A_\alpha|_e \geq \alpha |A_\infty|$ for all $\alpha > 0$ which is impossible unless $|A_\infty|_e = 0$.

We now show that $|A_{\alpha,\beta}| = 0$.

By Vitali Covering Theorem for Lebesgue measure on $A_{\alpha,\beta}$, given any $\epsilon > 0$, there exists a finite collection of pairwise disjoint balls $\{B_i\}_{i=1}^N$ such that

$$\left| A_{\alpha,\beta} \setminus \bigcup_{i=1}^N B_i \right|_e < \epsilon, \ \sum_{i=1}^N |B_i| < |A_{\alpha\beta}|_e + \epsilon, \quad \text{and } |B_i| > \frac{1}{\beta} u(B^{(\beta)}).$$

Then

$$\begin{aligned} & |A_{\alpha\beta}| + \epsilon \\ &> \textstyle\sum_{i=1}^N |B_i| && \text{(Vitali Covering Theorem for Lebesgue measure)} \\ &> \tfrac{1}{\beta} \textstyle\sum_{i=1}^N \mu(B_i) > \tfrac{1}{\beta} \sum_{i=1}^N \mu(\dot{B}_i) && (B_i \in \{B_U^{(\beta)}\} \subseteq \{B^{(\beta)}\}) \\ &\geq \tfrac{\alpha}{\beta} \textstyle\sum_{i=1}^N |\dot{B}_i \cap A_{\alpha,\beta}|_e && \text{(by claim 0.4 as open ball } \dot{B}_i \text{ contains } \dot{B}_i \cap A_{\alpha,\beta} \subseteq A_{\alpha,\beta}) \\ &\geq \tfrac{\alpha}{\beta} \left| A_{\alpha,\beta} \cap \textstyle\bigcup_{i=1}^N \dot{B}_i \right|_e && \text{(subadditivity of Lebesgue outer measure)} \\ &\geq \tfrac{\alpha}{\beta} \left| A_{\alpha,\beta} \cap \textstyle\bigcup_{i=1}^N B_i \right|_e && (|B_i \setminus \dot{B}_i| = 0) \\ &> \tfrac{\alpha}{\beta} |A_{\alpha,\beta}|_e - \tfrac{\alpha}{\beta}\epsilon && \text{(Vitali Covering Theorem for Lebesgue measure)} \end{aligned}$$

As ϵ to 0^+ is arbitrary, $|A_{\alpha,\beta}|_e \geq \frac{\alpha}{\beta} |A_{\alpha,\beta}|_e$. Hence $|A_{\alpha,\beta}| = 0$ as $\alpha > \beta$.

Remark 0.5.

- *Even though we only prove it for balls/cubes with center x that contract to x, we see that the same proof works for balls/cubes that contain x and contract to x.*

- *Instead of assuming* $u : \Sigma \to [0, \infty]$ *we could also consider additive set function* $u : \Sigma \to (-\infty, \infty)$*. All we need to do is to decompose it as a difference of two nonnegative additive set functions.*
- *If* u *is the Lebesgue measure, then* $Du(x) = 1$ *everywhere.*
- *If* $f : \mathbb{R}^n \to \mathbb{R}$ *is locally integrable and* $u(E) = \int_E f$*, then we have the Lebesgue Differentiation Theorem.*

In conclusion, we have the following theorem.

Theorem 0.6. *Given a set function* $u : \Sigma \to [0, +\infty]$ *that satisfies assumption (1) additivity and (2) monotoncity. Then*

$$Du(x) = \lim_{B \searrow x} \frac{u(B)}{|B|}$$

is defined for almost every x *where* $B \searrow x$ *denotes the contraction of closed balls* $\overline{B}_r(x), r > 0$ *into a singleton* x*.*

We next discuss absolute Continuous Set Functions.

Definition 0.7. *A set function* $u : \Sigma \to [0, +\infty]$ *(we now assume* Σ *is a* σ *algebra of measurable sets) is said to be absolute continuous if given any* $\epsilon > 0$*, there exists a* $\delta > 0$ *such that if* $E \in \Sigma$*,* $|E| < \delta$*, then* $u(E) < \epsilon$

Lemma 0.8 (Theorem 7.28)**.** *If* $u : \Sigma \to (-\infty, +\infty)$ *is an absolute continuous set function,* $E \in \Sigma$ *with* $|E| < +\infty$ *and* $\lim_{B \searrow x} \frac{u(B)}{|B|} = 0$ *for almost every* $x \in E$*, then* $u(E) = 0$*.*

For each $x \in E$, $\lim_{B \searrow x} \frac{u(B)}{|B|} = 0$, that is, given any $\epsilon > 0$, then there exists a $\delta > 0$ such that for all ball B with radius less than δ, i.e. $r(B) < \delta$, then $u(B) < \epsilon |B|$. Fix $\epsilon > 0$, there is a collection of balls $\{B^{(\epsilon)}\}$ that finely covers E and $u(B) < \epsilon |B|$ for all $B \in \{B^{(\epsilon)}\}$. By Vitali Covering Theorem for Lebesgue measure on finite measure E with $\{B^{(\epsilon)}\}$, given any $\epsilon, \delta > 0$, there exists a finite collection of pairwise disjoint balls $\{B_i\}_{i=1}^N$ such that $|E \setminus \bigcup_{i=1}^N B_i| < \delta$ and $\sum_{i=1}^N |B_i| < |E| + \epsilon$. Hence,

$$
\begin{aligned}
&|u(E)| \\
&= |u\left(E \setminus \bigcup_{i=1}^N B_i\right) + u\left(E \cap \bigcup_{i=1}^N B_i\right)| && \text{(additivity)} \\
&\leq \epsilon + |u\left(E \cap \bigcup_{i=1}^N B_i\right)| && \text{(by assumption)} \\
&\leq \epsilon + |u\left(\bigcup_{i=1}^N B_i\right)| && \text{(monotoncity of } |u|) \\
&= \epsilon + |\textstyle\sum_{i=1}^N u(B_i)| \leq \epsilon + \sum_{i=1}^N |u(B_i)| && \text{(additivity and triangle inequality)} \\
&\leq \epsilon + \epsilon \textstyle\sum_{i=1}^N |B_i| && (B_i \in \{B^{(\epsilon)}\}) \\
&\leq \epsilon + \epsilon(|E| + \epsilon) && \text{(Vitali Covering Theorem)}
\end{aligned}
$$

As $|E| < +\infty$, taking ϵ to 0^+ implies $u(E) = 0$.

Remark 0.9. *Put* $u([a,b]) = |f(b) - f(a)|$, $\lim_{B \searrow x} \frac{u(B)}{|B|} = 0$ *implies* f *being singular, recover* ***Theorem 7.28***

Theorem 0.10. *If* $u : \Sigma \to (-\infty, \infty]$ *is an absolute continuous set function. Given* $E \in \Sigma$ *with* $|E|_e < +\infty$*, given necessary conditions (additivity, subset) then* $u'(x) = \lim_{B \searrow x} \frac{u(B)}{|B|}$ *is defined almost every* $x \in E$*. Furthermore,*

$$u(E) = \int_E u'$$

for all $E \in \Sigma$*.*

u' is defined almost every x in a finite outer measure E is immediate from theorem 0.6. For the second part, let $v(E) = u(E) - \int_E u'$ for all $E \in \Sigma$, then

$$\lim_{B \searrow x} \frac{v(E)}{|E|} = \lim_{B \searrow x} \frac{u(E) - \int_E u'}{|E|} = \lim_{B \searrow x} \frac{u(E)}{|E|} - \lim_{B \searrow x} \frac{\int_E u'}{|E|} = 0 \text{ for almost every } x \in E$$

By Lemma 0.8, $v(E) = 0$ almost every where.

Remark 0.11. *Put* $u([a,b]) = f(b) - f(a)$*,* $f' = u' = \lim_{B \searrow x} \frac{u(B)}{|B|}$*, recover* ***Theorem 7.29***